Chapter 4 solutions

CH04-E1

1. Solve

$$x_1 + 2x_2 = 0, \qquad x_2 + x_3 = 0.$$

Set $x_2 = t$, giving $x_1 = -2t, x_3 = -t$. Hence

$$N(A) = \operatorname{span}\{(-2, 1, -1)^T\}.$$

2. The independent rows $r_1 = (1, 2, 0)^T, r_2 = (0, 1, 1)^T$ form a row-space basis. For $n = (-2, 1, -1)^T$,

$$r_1^T n = -2 + 2 = 0, \qquad r_2^T n = 1 - 1 = 0.$$

Every row-space combination is therefore perpendicular to n. The orthogonal spaces have dimensions 2 and 1, summing to 3, so they are orthogonal complements.

3. The 2×3 matrix has two independent rows and rank 2. Its column space is a two-dimensional subspace of $\mathbb{R}^2$, hence all of $\mathbb{R}^2$. The left nullity is

$$\dim N(A^T) = m - r = 2 - 2 = 0,$$

Thus $N(A^T) = \{0\}$.

Lin Cing https://github.com/xinhi6716/computer-science-self-study

CH04-E2

1. Since $a^T a = 3$ and $a^T b = 7$,

$$p = \frac{7}{3} a = \frac{7}{3}(1, 1, 1)^T.$$

2.

$$e = b - p = \left(2 - \frac{7}{3}, 1 - \frac{7}{3}, 4 - \frac{7}{3}\right)^T = \frac{1}{3}(-1, -4, 5)^T.$$

3. $a^T e = (-1 - 4 + 5)/3 = 0$, proving orthogonality to the line.

4. The line projector is

$$P = \frac{a a^T}{a^T a} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

It is symmetric. The product $J_3 J_3$ has every entry equal to 3, so

$$P^2 = \frac{1}{3^2} \cdot 3\, J_3 = \frac{1}{3} J_3 = P,$$

where J_3 is the all-ones matrix.

Chapter 4 solutions (continued)

CH04-E3

1.

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad x = \begin{bmatrix} c \\ m \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

2.

$$A^T A = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}, \qquad A^T b = \begin{bmatrix} 5 \\ 6 \end{bmatrix}.$$

The equations are

$$3c + 3m = 5, \qquad 3c + 5m = 6.$$

Subtraction gives $2m = 1$, hence $m = 1/2$, and substitution gives $c = 7/6$.

3.

$$p = Ax = \left(\frac{7}{6}, \frac{7}{6} + \frac{3}{6}, \frac{7}{6} + \frac{6}{6}\right)^T = \left(\frac{7}{6}, \frac{5}{3}, \frac{13}{6}\right)^T.$$

$$e = b - p = \left(-\frac{1}{6}, \frac{1}{3}, -\frac{1}{6}\right)^T.$$

The residual sum is zero, establishing orthogonality to the constant column. Also

$$t \cdot e = 0 + \frac{1}{3} + \left(-\frac{1}{3}\right) = 0.$$

Hence $A^T e = 0$.

4. $\|e\|^2 = 2/36 + 4/36 = 1/6$, the minimum sum of squared residuals over all c, m.

CH04-E4

1. Since $A^T A$ is symmetric, its inverse is symmetric.

$$P^T = A((A^T A)^{-1})^T A^T = A(A^T A)^{-1} A^T = P.$$

Also

$$P^2 = A(A^T A)^{-1} A^T A(A^T A)^{-1} A^T = P.$$

2. The complement is symmetric and

$$(I - P)^2 = I - 2P + P^2 = I - P.$$

Thus it is an orthogonal projector. The cross product is

$$P(I - P) = P - P^2 = 0.$$

Its range is perpendicular to the range of P because $P^T(I - P) = 0$.

Chapter 4 solutions (continued)

3. If $A^TAz = 0$, multiply by z^T to get

$$0 = z^TA^TAz = \|Az\|^2.$$

Therefore $Az = 0$. Column independence gives $z = 0$, proving a trivial nullspace and invertibility of A^TA.

4. Dependent columns give nonzero z with $Az = 0$, hence

$$A^TAz = 0,$$

This nonzero nullspace vector proves singularity of A^TA.

CH04-E5

1. The norm of a_1 is $\sqrt{2}$, so

$$q_1 = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right)^T.$$

Since $q_1^Ta_2 = 1/\sqrt{2}$,

$$z_2 = a_2 - \frac{1}{2}(1,1,0)^T = \left(\frac{1}{2}, -\frac{1}{2}, 1\right)^T.$$

$$\|z_2\| = \sqrt{\frac{1}{4} + \frac{1}{4} + 1} = \sqrt{\frac{3}{2}},$$

$$q_2 = \frac{1}{\sqrt{6}}(1,-1,2)^T.$$

2. The Gram-Schmidt representations are

$$a_1 = \|a_1\|q_1 = \sqrt{2}\,q_1,$$

$$a_2 = \frac{1}{\sqrt{2}}q_1 + \sqrt{\frac{3}{2}}q_2.$$

Therefore

$$R = \begin{bmatrix} \sqrt{2} & 1/\sqrt{2} \\ 0 & \sqrt{3/2} \end{bmatrix}.$$

3. Both columns have unit norm, and

$$q_1^Tq_2 = \frac{1 + (-1) + 0}{\sqrt{12}} = 0,$$

Thus $Q^TQ = I_2$. Expanding by columns gives

$$\sqrt{2}q_1 = a_1,$$

$$\frac{1}{\sqrt{2}}q_1 + \sqrt{\frac{3}{2}}q_2 = \left(\frac{1}{2}, \frac{1}{2}, 0\right)^T + \left(\frac{1}{2}, -\frac{1}{2}, 1\right)^T = a_2,$$

Chapter 4 solutions (continued)

Hence $QR = A$.

CH04-E6

1.

$$Q^T b = \left(\frac{1+2}{\sqrt{2}}, \frac{1-2}{\sqrt{6}}\right)^T = \left(\frac{3}{\sqrt{2}}, -\frac{1}{\sqrt{6}}\right)^T.$$

Solve

$$\sqrt{2}\,x_1 + \frac{1}{\sqrt{2}}x_2 = \frac{3}{\sqrt{2}}, \qquad \sqrt{\frac{3}{2}}x_2 = -\frac{1}{\sqrt{6}}.$$

The second equation gives $x_2 = -1/3$; the first then gives $x_1 = 5/3$. Thus

$$x = \begin{bmatrix} 5/3 \\ -1/3 \end{bmatrix}.$$

2.

$$p = Ax = \left(\frac{4}{3}, \frac{5}{3}, -\frac{1}{3}\right)^T,$$

$$e = b - p = \frac{1}{3}(-1, 1, 1)^T, \qquad \|e\|^2 = \frac{1}{3}.$$

3.

$$a_1^T e = -\frac{1}{3} + \frac{1}{3} = 0,$$

$$a_2^T e = -\frac{1}{3} + \frac{1}{3} = 0.$$

This nonzero residual distinguishes a least-squares solution from an exact solution. Columnwise orthogonality is precisely $A^T e = 0$, and full column rank guarantees a unique parameter vector.

CH04-E7

1. The columns are identical, so they are dependent and rank is 1.
2. Set $a = (1, 2)^T$. Then

$$a^T a = 5, \qquad a^T b = 1 + 6 = 7,$$

$$p = \frac{7}{5}a = \left(\frac{7}{5}, \frac{14}{5}\right)^T.$$

3.

$$A^T A = \begin{bmatrix} 5 & 5 \\ 5 & 5 \end{bmatrix}, \qquad A^T b = \begin{bmatrix} 7 \\ 7 \end{bmatrix}.$$

Lin Cing https://github.com/xinhi6716/computer-science-self-study

Chapter 4 solutions (continued)

Every minimizer has the form

$$x^\star = \begin{bmatrix} 7/5 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \quad t \in \mathbb{R}.$$

4. Differences of parameters producing p lie in $N(A) = \operatorname{span}\{(-1,1)^T\}$. This nonzero direction gives nonunique parameters, while all map to the same nearest column-space point p.

CH04-E8

1. Direct multiplication gives

$$P^2 = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} = P.$$

2. However,

$$P^T = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} \neq P.$$

3. Since $P(x,y)^T = (x+y, 0)^T$,

$$R(P) = \operatorname{span}\{(1,0)^T\}, \qquad N(P) = \operatorname{span}\{(-1,1)^T\}.$$

4. $R(P)$ and $N(P)$ are not orthogonal, since $(1,0)^T \cdot (-1,1)^T = -1 \neq 0$. Thus $P^2 = P$ describes an oblique projector; orthogonal projection also requires $P^T = P$.

CH04-E9

The row space has basis $(1,1,0)^T$, the column space $(1,2)^T$, both of dimension 1. Solving Ax=0 gives basis $(1,-1,0)^T, (0,0,1)^T$, dimension 2. The left nullspace has basis $(2,-1)^T$, dimension 1. Their dot products verify the two orthogonal pairs, and dimensions sum to 3 on the input side and 2 on the output side.

Projection onto c=(1,2) gives $p = c(c^T b)/(c^T c) = (1,2)^T$, $e = (2,-1)^T$, with $c^T e = 0$. For any x, e is orthogonal to p−Ax, so

$$\|b - Ax\|^2 = 5 + \|p - Ax\|^2 \geq 5.$$

Equality holds exactly when Ax=p, or $x_1+x_2=1$. Hence all minimizers are $x = (1/2 + t, 1/2 - t, s)^T$. Split this into $\hat{x} = (1/2, 1/2, 0)^T$ and $z = (t, -t, s)^T$, in the orthogonal row and null spaces. Thus $\|x\|^2 = 1/2 + 2t^2 + s^2$, uniquely minimized at t=s=0, with norm $1/\sqrt{2}$.

Chapter 4 solutions (continued)

For the particular solution, $(2, -1, 2)^T = \hat{x} + (3/2, -3/2, 2)^T$. Its nullspace component maps to zero and has squared norm 17/2. Its total squared norm is 1/2+17/2=9, so its norm is 3. It has the same projected output p and minimum residual norm $\sqrt{5}$, but a larger parameter norm. Residual norm and parameter norm are distinct objectives.

Verification code appendix: projection_and_qr.py

```
"""Check the line projection and Gram-Schmidt example in chapter four."""

from math import sqrt


def dot(left, right):
    return sum(x * y for x, y in zip(left, right))


def scale(value, vector):
    return [value * x for x in vector]


def add(left, right):
    return [x + y for x, y in zip(left, right)]


def norm(vector):
    return sqrt(dot(vector, vector))


a = [1.0, 1.0, 1.0]
b = [2.0, 1.0, 4.0]
p = scale(dot(a, b) / dot(a, a), a)
e = add(b, scale(-1.0, p))
assert abs(dot(a, e)) < 1e-12
assert abs(norm(e) - sqrt(42) / 3) < 1e-12

a1 = [1.0, 1.0, 0.0]
```

Lin Cing https://github.com/xinhi6716/computer-science-self-study

Chapter 4 solutions (continued)

```
a2 = [1.0, 0.0, 1.0]
q1 = scale(1.0 / norm(a1), a1)
z2 = add(a2, scale(-dot(q1, a2), q1))
q2 = scale(1.0 / norm(z2), z2)
assert abs(dot(q1, q1) - 1.0) < 1e-12
assert abs(dot(q2, q2) - 1.0) < 1e-12
assert abs(dot(q1, q2)) < 1e-12

r11 = sqrt(2.0)
r12 = 1.0 / sqrt(2.0)
r22 = sqrt(1.5)
reconstructed_a1 = scale(r11, q1)
reconstructed_a2 = add(scale(r12, q1), scale(r22, q2))
assert all(abs(x - y) < 1e-12 for x, y in zip(reconstructed_a1, a1))
assert all(abs(x - y) < 1e-12 for x, y in zip(reconstructed_a2, a2))

target = a2
qtb = [dot(q1, target), dot(q2, target)]
x2 = qtb[1] / r22
x1 = (qtb[0] - r12 * x2) / r11
fit = add(scale(x1, a1), scale(x2, a2))
assert abs(x1) < 1e-12 and abs(x2 - 1.0) < 1e-12
assert norm(add(target, scale(-1.0, fit))) < 1e-12

print("line projection residual norm =", norm(e))
print("q1 =", q1)
print("q2 =", q2)
```

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

References and attribution

Gilbert Strang. Introduction to Linear Algebra, 5th Edition, 2016.

Original educational synthesis by Lin Cing. Original material is licensed under CC BY 4.0. The referenced textbook is not distributed with these notes.

https://github.com/xinhi6716/computer-science-self-study